Emotion Diary

DIPLOMA PROJECT

Student Name: Hanna Drozhdzh
Group: 22-HR-JS
Supervisor: Yahor Bialiauski
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PROBLEM STATEMENT

Problem:

- Young adults can’t identify, label, or understand their emotions.
- Emotional intelligence rarely taught in schools/workplaces.

Low emotional intelligence leads to:

- Decreased well-being,
- Increased anxiety and depression,
- Difficulty navigating personal and professional relationships
- Reduced ability to cope with life stressors.

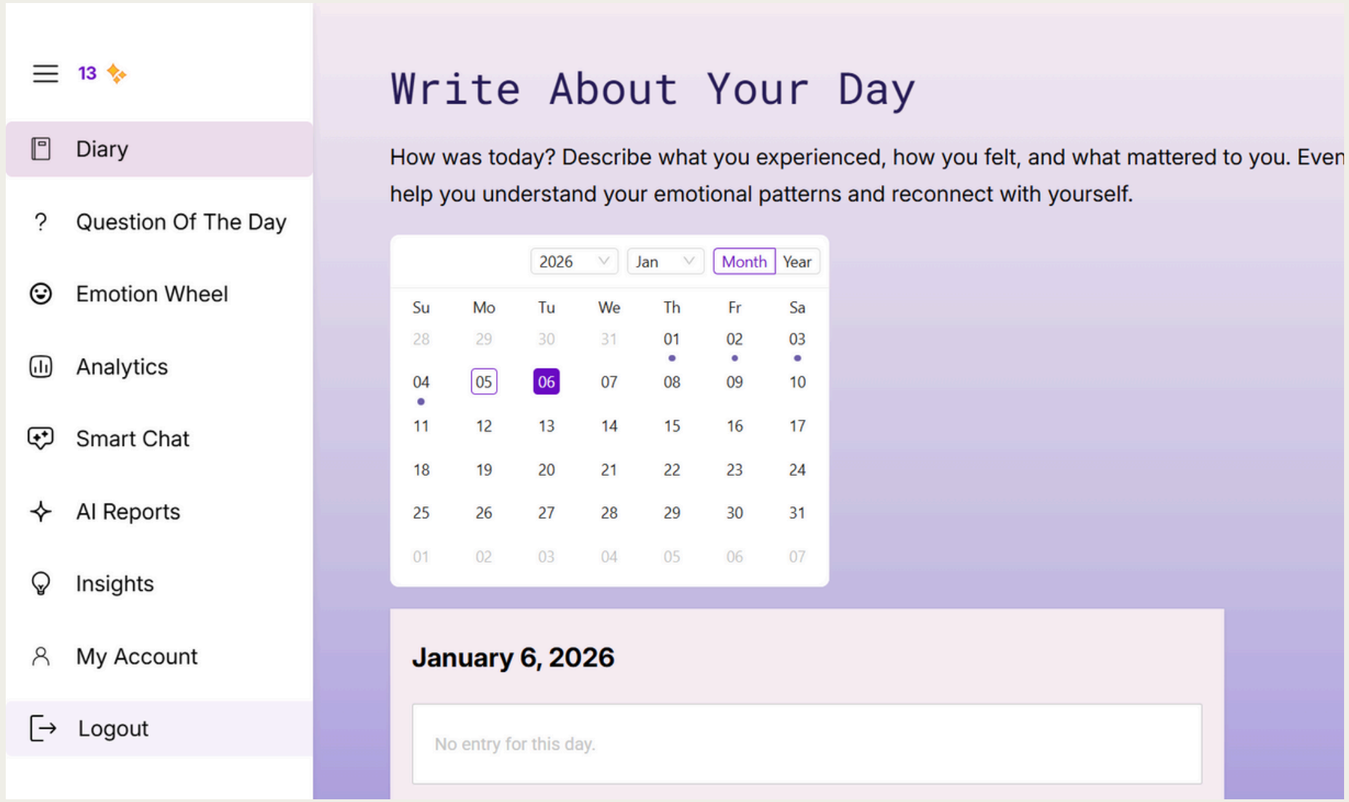
Target Users:

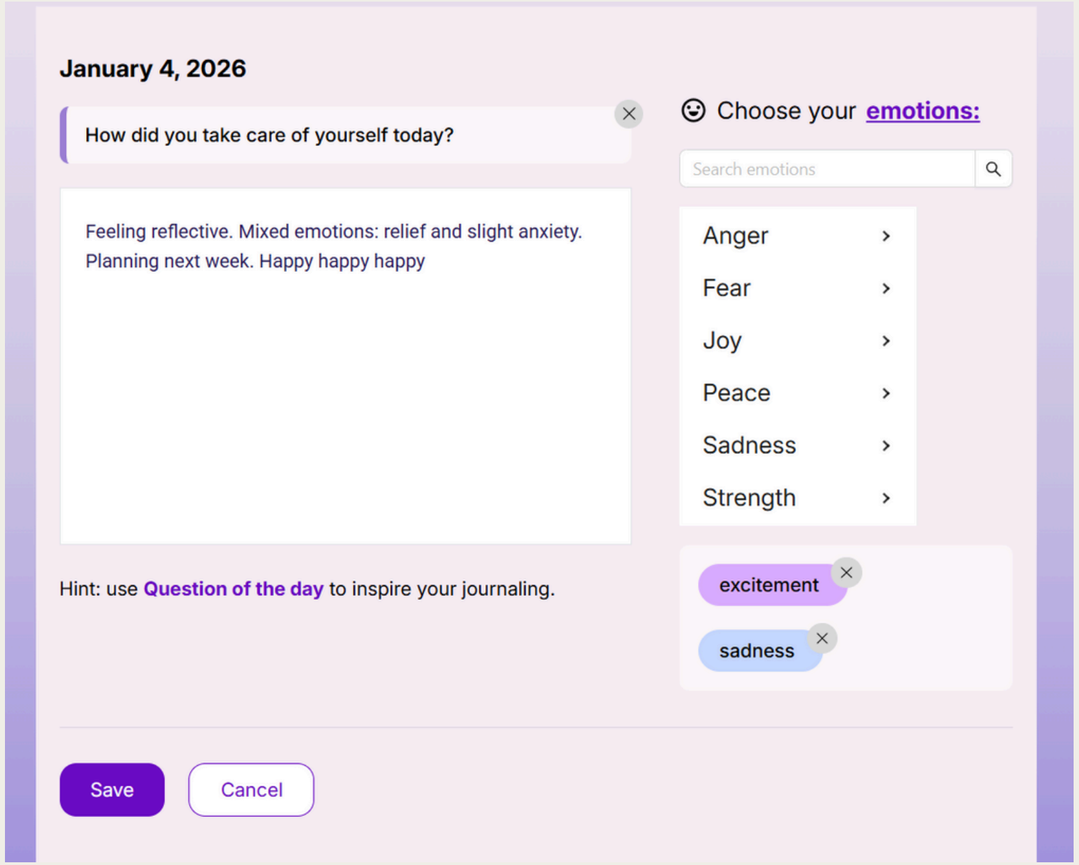
- Students, freelancers, early-career professionals (17–35).

EMOTION DIARY: KEY FEATURES

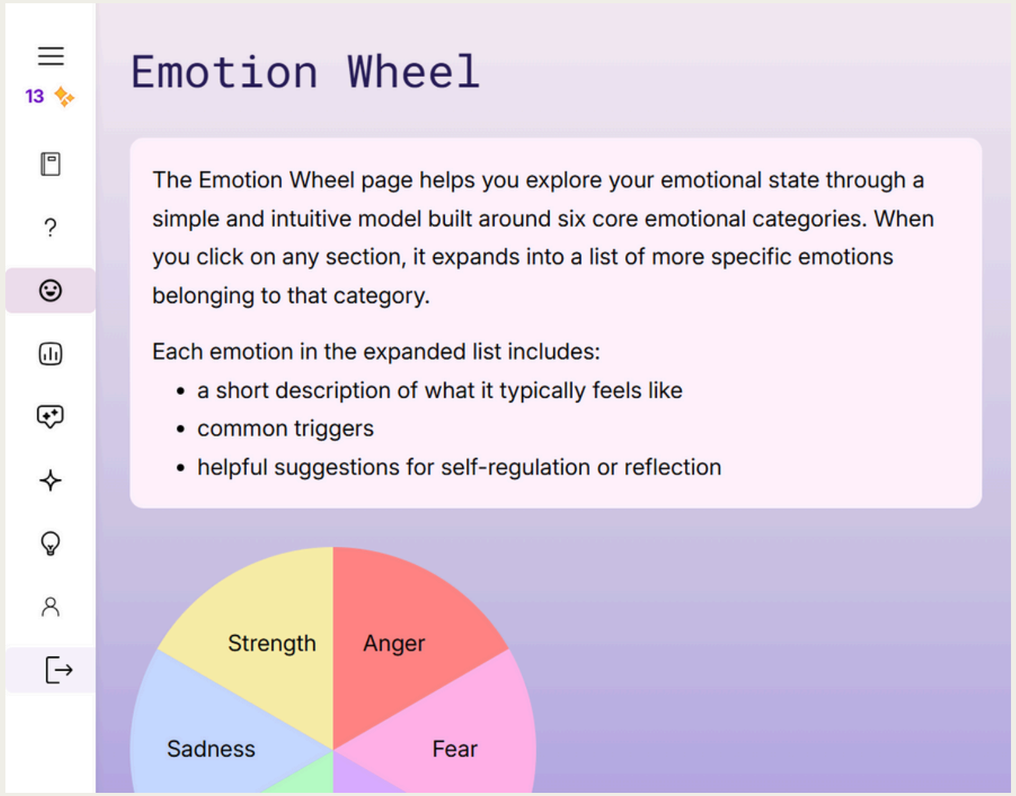
1. **Track & Reflect Emotions**
   - Daily journaling with emotion tagging → helps users name and recognize feelings.
2. **AI-Powered Insights**
   - Detects patterns, triggers, and provides actionable recommendations → users understand why they feel in a certain way.
3. **Educational Support**
   - Emotion wheel → teach emotional literacy and self-regulation.
4. **Motivation & Consistency**
   - Gamified streaks and personalized feedback → encourage daily journaling habits.

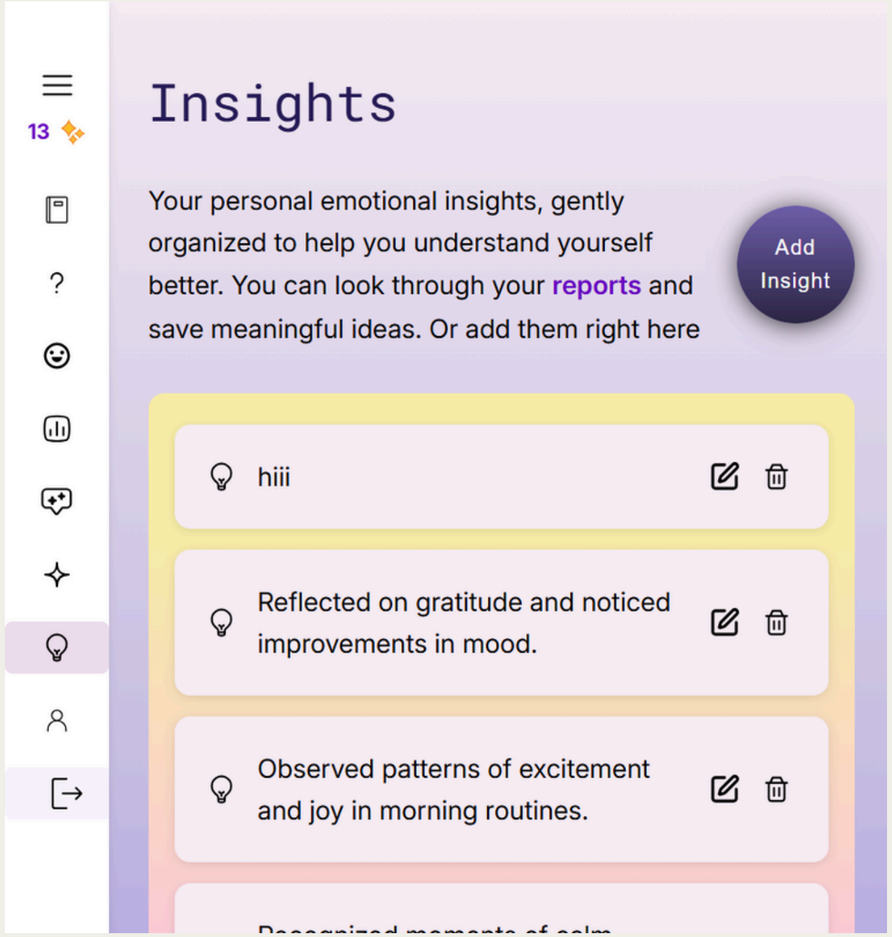
SCREENSHOTS

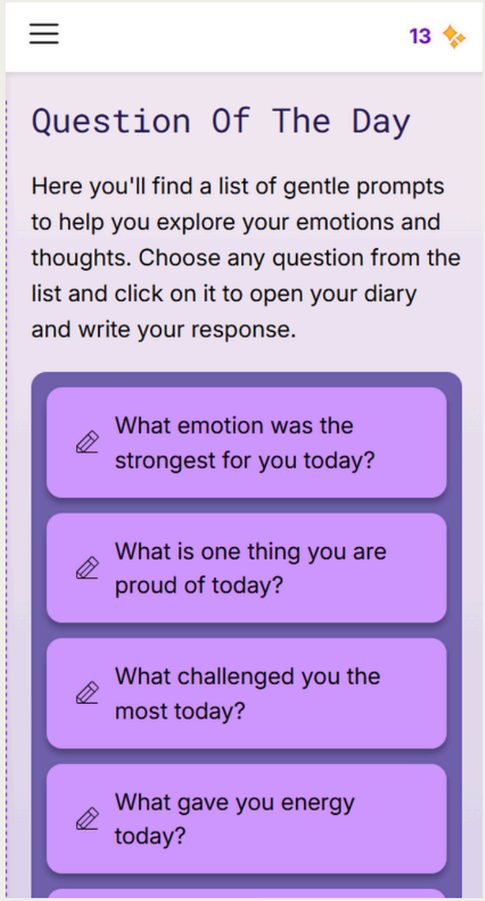




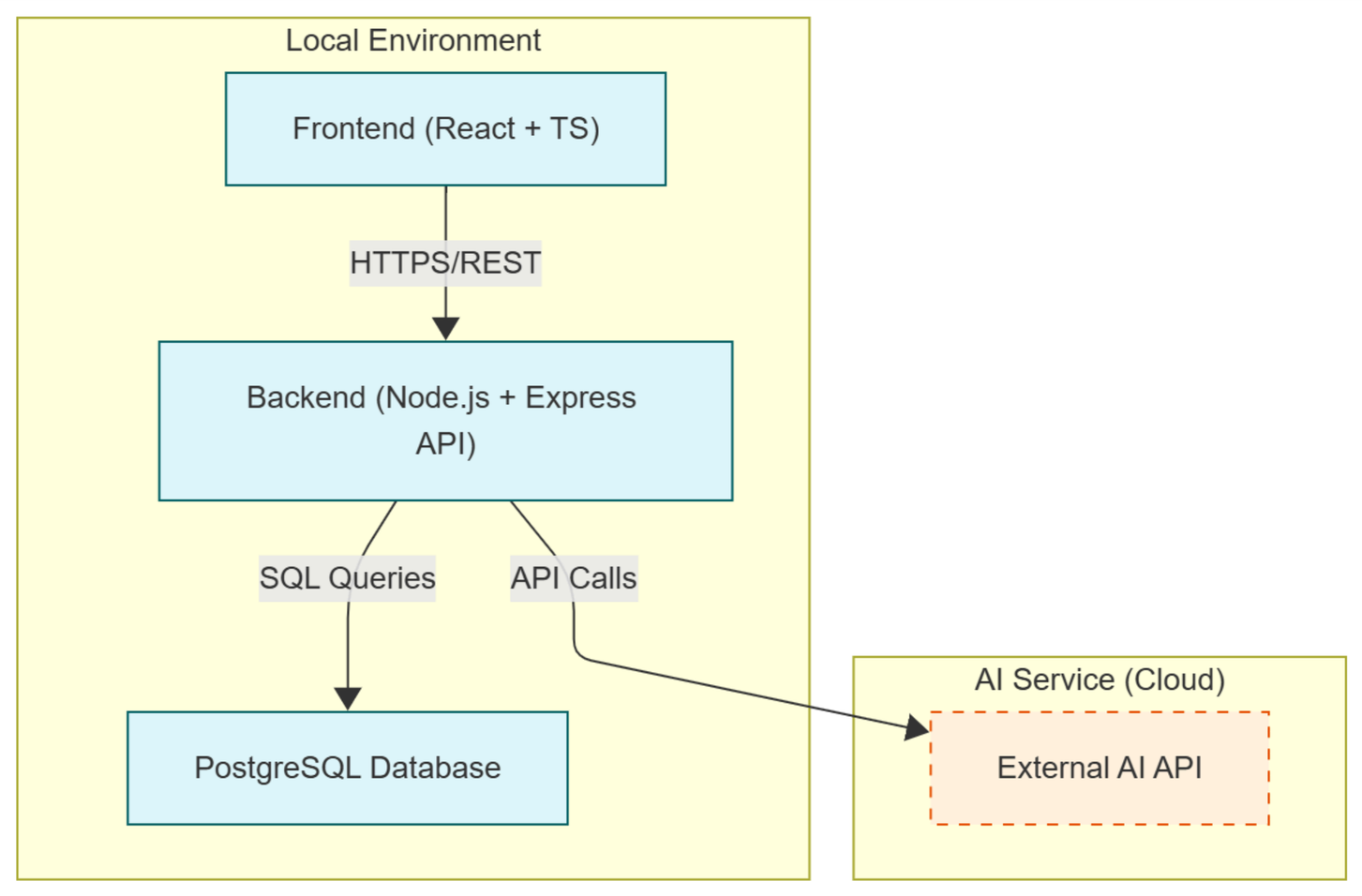
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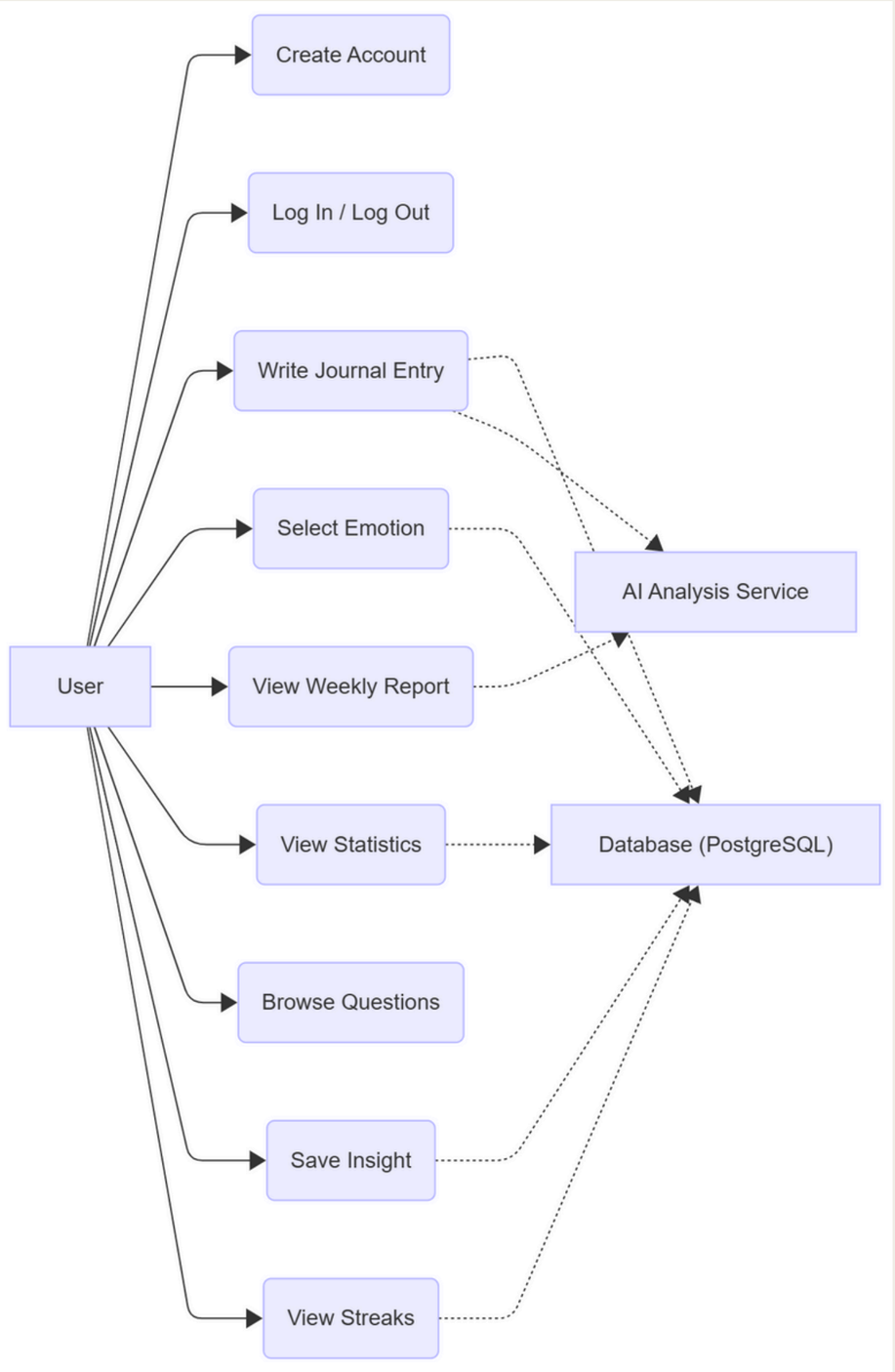
ARCHITECTURE



Technology Stack

- Frontend: React 19, TypeScript, Ant Design
- Backend: Node.js 20, Express.js, TypeScript
- Database: PostgreSQL 16
- AI Integration: Google Gemini API
- Deployment: Vercel (Frontend), Railway (Backend + DB)
- Containerization: Docker + Docker Compose

USE CASE DIAGRAM



C R I T E R I A

1.Frontend

- SPA
- React 19 + TypeScript
- Ant Design UI + SCSS
- Component-based architecture, protected routing

2.Backend

- Node.js 20 + Express.js + TypeScript
- Layered architecture (Routes – Controllers – Services – Repositories)
- JWT authentication, API documentation (Swagger)

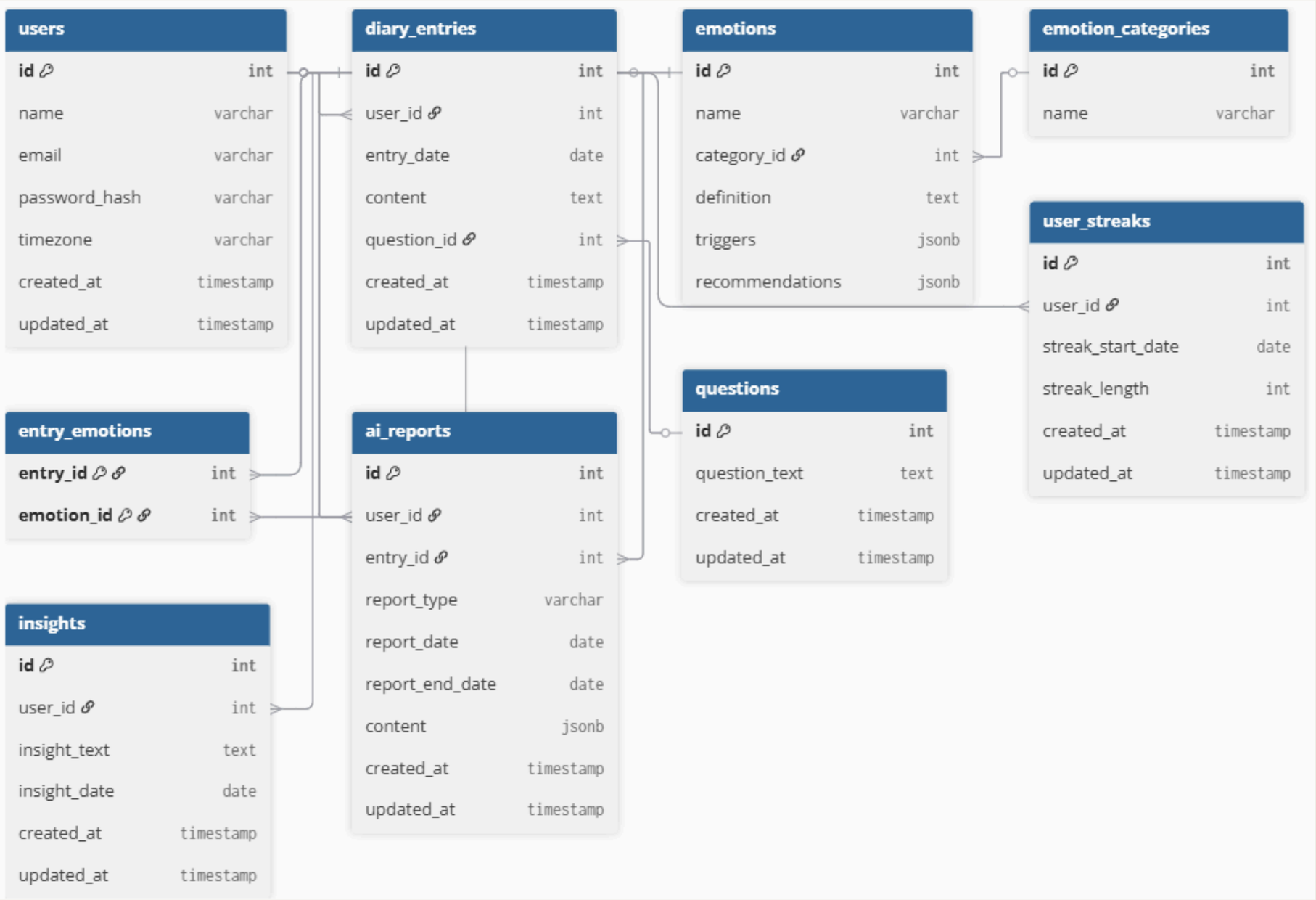
3. Database

- PostgreSQL 16, relational database (10 tables, 3NF)
- Role-based access control
- Migrations and automated triggers

4.AI Assistant

- Emotion analysis from journal entries
- Weekly pattern detection and empathetic chat
- Google Gemini 2.5 Flash

DATABASE SCHEMA



CRITERIA

5.Refined UX

- Guided journaling and emotion visualization
- Clear navigation and empty states
- Calm, accessible design (WCAG 2.1)

6.Adaptive UI

- Mobile, tablet, and desktop support
- Responsive breakpoints and adaptive layouts
- Touch-friendly components

7.Containerization

- Full containerization with Docker
- Docker Compose for service orchestration
- Separate development and production environments

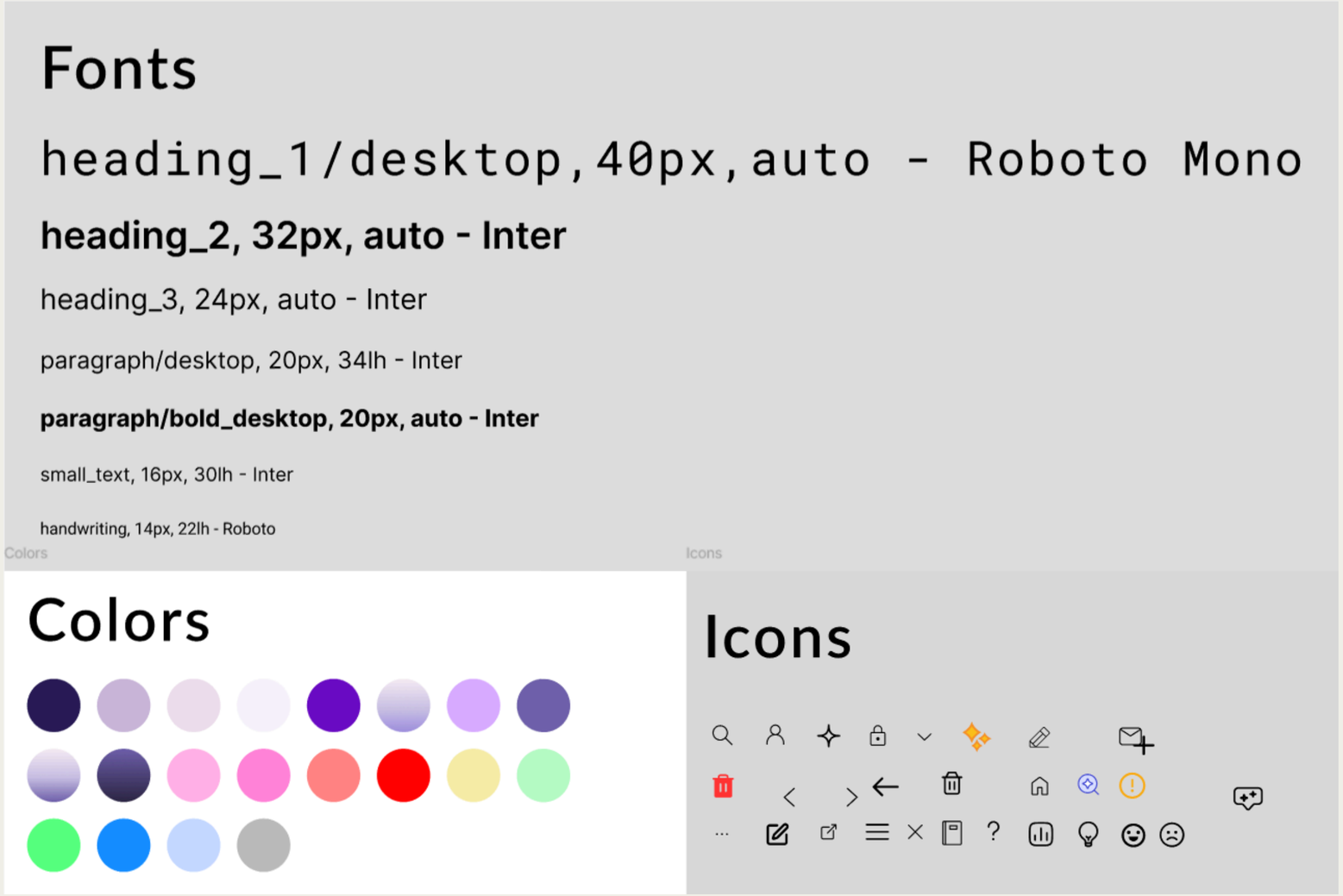
8.Testing

- Qualitative UX testing
- Heuristic evaluation and scenario-based testing
- 4 participants, 19 test cases, 5 UX issues and 1 bug fixed

DESIGN SYSTEM

Figma design:

https://www.figma.com/design/xgrs1dQC3LwKxozlFfmQ17/Emotion?node-id=0-1&p=f&t=wfRWmRm6fairgIk4-0



CHALLENGES & SOLUTIONS

1. Challenge: First-time production deployment

Solution: Dockerized app, tested locally, step-by-step Railway & Vercel setup

Lesson: Clear startup commands, well-defined environment variables, and documented deployment steps are critical for a stable and reproducible production setup

2. Challenge: Inconsistent AI JSON responses

Solution: Implemented a JSON cleanup and validation layer before processing AI responses

Lesson: AI output must always be treated as untrusted input and validated explicitly

RESULTS

✅ Completed Features (MVP)

- User authentication (JWT)
- Journal entries (create, edit, delete, calendar view)
- Emotion tracking with predefined emotion set
- AI Integration (chat, daily and weekly reports)
- Analytics, Question of the day, Insights, Emotion Wheel functionality
- Responsive UI (mobile / tablet / desktop)
- PostgreSQL database with migrations, triggers, and views
- Dockerized full-stack application
- Production deployment (Railway + Vercel)
- User Research and Qualitative Testing

⚠️ Partially Implemented / Known Issues

- Streaks & gamification
- Analytics edge cases

🧩 Future Backlog

- Automated testing (unit, integration, E2E)
- Data export (JSON / PDF)
- Notifications (email / push)
- Mobile-native apps (React Native)

DEMO

L I N K S

Emotion Diary URL: https://emotion-amber.vercel.app

Repository link: https://github.com/drnyuta/Emotion/tree/main

Design link (Figma): https://www.figma.com/design/xgrs1dQC3LwKxozlFfmQ17/Emotion?node-id=1-2&p=f&t=k4u5GRO9u3NbZ1pP-0

Github

Web application

Thank you!

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